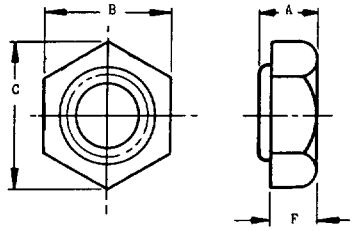
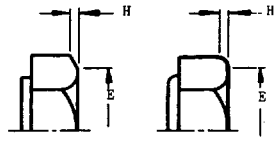
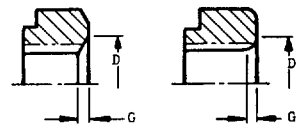
FED. SUP CLASS
5310




CONFIGURATION OF NUT IS OPTIONAL, LIMITATIONS IMPOSED ONLY ON DIMENSIONS AND REQUIREMENTS SPECIFIED HEREIN




WASHER FACE, CHAMFER OR RADIUS RELIEF TO HEX OPTIONAL WHEN WITHIN THE LIMITATIONS OF DIMENSIONS E AND H




WASHER FACE, CHAMFER OR RADIUS RELIEF TO THREAD OPTIONAL WHEN WITHIN THE LIMITATIONS OF DIMENSIONS D AND G

| SIZE DASH NO. | (a) THREAD | A | | B | | C REF | D DIA | | E MIN DIA | F MIN | G | | H MAX | AXIAL STRENGTH LBS MIN | WEIGHT LBS/100 MAX |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAX | MIN | MAX | MIN | | MAX | MIN | | | MAX | MIN | | | |
| 04 | 4-40 NC-3B | .125 | .112 | .252 | .240 | .289 | .154 | .112 | .190 | .081 | .046 | .039 | .024 | — | .13 |
| 06 | 6-32 NC-3B | .141 | .126 | .314 | .302 | .361 | .178 | .138 | .252 | .092 | .049 | .042 | .027 | — | .23 |
| 08 | 8-32 NC-3B | .188 | .140 | .316 | .334 | .390 | .203 | .173 | .284 | .122 | | | .037 | 1,720 | .28 |
| 3 | 10-32 NF-3B | | .147 | .377 | .365 | .430 | .230 | .199 | .315 | | | | | 2,460 | .35 |
| 4 | 1/4 -28 UNF-3B | .219 | .177 | .439 | .430 | .506 | .293 | .259 | .380 | .142 | .055 | .047 | .042 | 4,580 | .53 |
| 5 | 5/16-24 UNF-3B | .266 | .239 | .502 | .492 | .580 | .356 | .322 | .452 | .173 | .059 | .051 | .052 | 7,390 | .85 |
| 6 | 3/8 -24 UNF-3B | .282 | .269 | .564 | .553 | .650 | .418 | .385 | .523 | .183 | | | .055 | 11,450 | 1.10 |
| 7 | 7/16-20 UNF-3B | .328 | .315 | .690 | .679 | .790 | .482 | .448 | .629 | .213 | .066 | .056 | .064 | 15,540 | 1.95 |
| | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | |

(a) THREADS IN ACCORDANCE WITH MIL-S-7742, BEFORE LUBRICATION.

MATERIAL: STEEL, SEE PERFORMANCE SPECIFICATION.
PLATING: CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 3.
LUBRICANT: DRY FILM, WHERE SPECIFIED BY THE CODE LETTER L.
FOR USAF APPLICATIONS, NUTS TREATED WITH DRY FILM LUBRICANTS SHALL NOT BE UTILIZED IN INTEGRAL FUEL TANKS.

DIMENSIONS IN INCHES.

LETTER L BEFORE FIRST DASH NUMBER DESIGNATES DRY FILM LUBRICANT COATED NUTS.

EXAMPLES OF PART NUMBER: MS21040-4 = 1/4-28 NUT, ALL METAL, CADMIUM PLATED.
MS21040L4 = 1/4-28 NUT, ALL METAL , CADMIUM PLATED, DRY FILM LUBRICATED.

ONLY NUTS FOR WHICH THERE ARE QUALIFIED PRODUCTS LISTED ON QPL25027 SHALL BE USED FOR DESIGN.
DESIGN INFORMATION: THESE NUTS ARE DESIGNED TO DEVELOP THE TENSILE STRENGTH OF BOLTS AND SCREWS WITH AN ULTIMATE TENSILE STRENGTH OF 125,000 PSI BASED ON THE AREA AT THE BASIC ROOT DIAMETER OF THE BOLT OR SCREW THREAD.

INTERCHANGEABILITY RELATIONSHIP: MS21040 NUTS CAN UNIVERSALLY REPLACE AN363, MS20364, AND MS20365 NON-CORROSION RESISTANT STEEL NUTS OF LIKE THREAD SIZE, AND WHICH ARE EITHER ALL METAL OR HAVE NON-METALLIC INSERTS: BUT THESE AN363, MS20364, AND MS20365 NUTS CAN NOT UNIVERSALLY REPLACE THE MS21040 NUTS.

PROVISIONS OF THIS STANDARD ARE THE SUBJECT OF INTERNATIONAL STANDARDIZATION AGREEMENT. WHEN REVISION OR CANCELLATION OF THIS STANDARD IS PROPOSED, THE DEPARTMENTAL CUSTODIANS WILL INFORM THEIR RESPECTIVE DEPARTMENTAL STANDARDIZATION OFFICES SO THAT APPROPRIATE ACTION MAY BE TAKEN RESPECTING THE INTERNATIONAL AGREEMENT CONCERNED.

## INACTIVE FOR DESIGN AFTER 23 SEPTEMBER 1960. USE MS21042.

INTERCHANGEABILITY RELATIONSHIP: MS21042 NUTS CAN UNIVERSALLY REPLACE THE INACTIVATED MS21040 NUTS OF LIKE THREAD SIZE AND LUBRICANT (DRY FILM AND NON-DRY FILM); BUT THE INACTIVATED MS21040 NUTS CANNOT ALWAYS REPLACE THE SUPERSEDING MS21042 NUTS.

SHEET NO. 2 DELETED.

## © CANCELLED AFTER 22 JULY 1969. USE MS21042. SEE INTERCHANGEABILITY TABLE ON MS21042 FOR SUBSTITUTIVE PARTS.

THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.
REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

| P.A. Navy-AS Other Cust USAF-11 Army-AV | INTERNATIONAL INTEREST | TITLE: NUT, SELF-LOCKING, 500°F, HEXAGONAL, REDUCED HEIGHT | MILITARY STANDARD MS21040 |
|---|---|---|---|
| PROCUREMENT SPECIFICATION MIL-N-25027 | | SUPERSEDES: | SHEET 1 OF 1 |

DD FORM 1 SEP 67 672-1 (Limited coordination) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

User activities: Army-EL
Review activities: Navy-AS, WC, MI, AS Army-AV, USAF-11, 82
DSA-IS

REVISED (A) 23 Sep 60 (B) 29 Jun 62 (C) 22 Jul 69
APPROVED 25 May 59

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

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